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ENVIRONMENTAL BIOTECHNOLOGY

Regulatory mechanism of montmorillonite on antibiotic resistance genes in *Escherichia coli* induced by cadmium

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Abstract
The emergence and spread of antibiotic resistance genes (ARGs) induced by the overuse of antibiotics has become a serious threat to public health. Heavy metals will bring longer-term selection pressure to ARGs when the concentration of their residues is higher than that of antibiotics in environmental media. To explore the potential roles of montmorillonite (Mt) in the emergence of ARGs under divalent cadmium ion (Cd^{2+}) stress, *Escherichia coli* (*E. coli*) was induced continuously for 15 days under Cd^{2+} gradient concentrations (16, 32, 64, 96, and 128 μg•mL^{-1}) with and without Mt. Subsequently, antibiotic resistance testing, transcriptomics, transmission electron microscope, scanning electron microscopy, and Fourier transform infrared were conducted for analysis. The results of characterization analysis showed that Cd^{2+}could enhance the expression of resistance genes such as penicillin, tetracycline, macrolactone, and chloramphenicol in *E. coli*. Moreover, compared with Cd^{2+}, Mt-Cd could inhibit the promotion of these resistances by alleviating the expressions of genes involved in cell wall/membrane, protein synthesis, transport systems, signal transduction, and energy supply processes. Therefore, the study promoted the understanding of Cd^{2+} in triggering bacterial antibiotic resistance and highlighted a novel theme of clay's ability to mitigate ecological risk of antibiotic resistance caused by heavy metals.

Key points
- *Montmorillonite (Mt) could inhibit the promotion of antibiotic resistances.*
- E. coli formed a unique resistance mechanism by interacting with Mt and Cd^{2+}.
- *Mt stimulated cellular signal transduction, cellular component, and energy supply.*

Keywords Antibiotic resistance genes · Escherichia coli · Cadmium · Montmorillonite

Introduction

Antibiotics, as bactericidal drugs, are widely used in clinical, agricultural, and aquaculture fields (Zhou et al. 2013), which has greatly promoted the development of human society. However, overuse of antibiotics brought about the intergenerational transmission and spread of antibiotic resistance, resulting in the emergence of strains carrying antibiotic resistance genes (ARGs), which was a threat to public health (Leonard et al. 2018).

Cadmium (Cd), which ascribed to its high toxicity and carcinogenesis even at relatively low doses, is considered

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as one of the five most hazardous elements in the environment (Xie et al. 2018). The contamination with cadmium is of great concern all around the world, especially in China; according to a study of 146 cities in China by Wang et al. (2015), Cd contamination level was higher than the level in 1990, and the Cd contamination of soil in urban areas was more serious than contamination in agricultural areas. Besides, Cd in the soil had strong metabolic activity, which could easily cause microbial resistance (Wang et al. 2020). Crucially, heavy metals were not degradable, so they imposed a longer term selection pressure on ARGs compared to antibiotics (Stepanauskas et al. 2005). Therefore, it is necessary to explore the effect of Cd on the expression of ARGs. In recent year, a growing number of studies have presented those non-antibiotic substances, such as heavy metals could also significantly induce the production and transmission of ARGs (Knapp et al. 2011). For example, Wei et al. (2020) found that the relative abundance of target ARGs increased under the influence of Cd^{2+} in bulk and rhizosphere soils and endophytes. Wu et al. (2015) reported that the abundance of resistance genes such as *sul1*, *sul2*, *tetM*, *tetQ*, *ermB*, and *mefA* was positively correlated with the content of Cr, Cd, Ni, and As. However, the molecular regulation mechanism of heavy metal–induced antibiotic resistance genes expression is still poor.

Montmorillonite (Mt) was the typical clay mineral with abundant functional groups of Al–OH/Si–OH, which was widely distributed in soil and groundwater environment (Hu et al. 2020). It has a large specific surface area and can be bound with hydrated exchangeable cations between the cations and various heavy metal ions (Chen et al. 2020). Besides, Mt can alter environmental conditions by adjusting pH or retaining moisture, improving the adaptability of surrounding microorganisms to harsh environmental conditions (Ruan et al. 2018; Wong et al. 2004). Recently, various attempts have been made to explore the relationship between ARGs and clays. Some researchers reported that clays could promote transfer of ARGs by influencing the permeability of bacterial cell membranes or plasmid (Li et al. 2020; Wang et al. 2018). In addition, our previous study showed that kaolinite could inhibit the production of ARGs under the stress of ampicillin (Lai et al. 2019). The coupled reaction of microbes and Mt would increase bacterial exposure to Mt-adsorbed Cd^{2+}, which further enhances interactions between bacteria and Cd^{2+} (Wang et al. 2020). These studies showed that clays had the properties of regulating bacterial metabolism and relieving environmental stress. However, whether Mt can influence the emergence of bacterial ARGs under Cd^{2+} stress is unclear. Hence, it is of great significance for us to better understand the expression of antibiotic resistance genes in bacteria to reveal the regulatory mechanism of interface reaction among montmorillonite, Cd^{2+}, and bacteria.

Escherichia coli (*E. coli*) has been widely studied for its susceptibility to multiple antibiotic resistance (Wuijts et al. 2017). Thus, in this paper, wild-type *E. coli* was exposed to Cd^{2+}-containing medium in the presence and absence of Mt at gradient concentrations for multi-step induction. Transcriptome approaches, phenotypic test, and characterization profiling were conducted to investigate the regulation mechanism of Mt on bacterial ARGs after 15-day Cd^{2+} exposure. The purpose of this study was expected to help us better understand the regulatory mechanism of ARGs expression under the influence of clay and heavy metals and provide a new perspective for the treatment of antibiotic pollution.

Materials and methods

Reagents, strains, and culture conditions

$Cd(NO_3)_2 \cdot 4H_2O$ was purchased from Aladdin Industrial Company. Tetracycline was purchased from Sigma Aldrich Chemical Company. Ampicillin, erythromycin, chloramphenicol, Mueller Hinton broth (MHB), and Luria–Bertani (LB) medium (10 $g{\bullet}L^{-1}$ sodium chloride, 10 $g{\bullet}L^{-1}$ tryptone, 5 $g{\bullet}L^{-1}$ yeast extract) were purchased from Beijing Solarbio Science & Technology Company. All the chemicals were of analytical grade, and all experiments were done with deionized (DI) water. Wild *E. coli* ATCC25922 strain used in the study was purchased from China General Microbiological Culture Collection. It was shaken overnight in LB medium at 37 °C, 200 rpm, and pH 7.0.

Design of induction experiment

In this study, Cd^{2+} gradient concentration induction method was adopted to rapidly induce antibiotic-sensitive *E. coli* to increase antibiotic resistance by gradually increasing Cd^{2+} concentration based on its sub-MIC (minimum inhibitory concentration). The Mt purchased from Feilaifeng Non-metallic Minerals Material Company was washed 3 times and dried at 60 °C. Then, it was passed through a 200-mesh sieve and finally stored in a sealed bag with desiccator for subsequent experimental analysis. First, 16 $g{\bullet}L^{-1}$ Mt was added to Erlenmeyer flask for sterilization, and then quantitative LB medium was added. Bacterial turbidimeter (Qi Wei Instrument Co. Ltd., Hangzhou) was used to adjust the purified *E. coli* to 2.5 MCF (1 MCF $= 3 \times 10^8$ $CFU{\bullet}mL^{-1}$) as bacterial suspension. The suspension was added at a ratio of 1:100 to a fresh LB medium with or without 16 $g{\bullet}L^{-1}$ Mt for three generations (each generation referred to overnight culture). The initial Cd^{2+} concentration of 16 $\mu g{\bullet}mL^{-1}$ in LB was adopted based on MIC (32 $\mu g{\bullet}mL^{-1}$) determined before the induction experiment (Supplementary information). The culture conditions were 37 °C, pH 7.0, and a speed of 200 rpm. Next, under the same conditions, strains were transferred to the next Cd^{2+} medium with a concentration gradient (32, 64, 96, and 128 μg•mL^{-1}) for 3 generations by the same way, respectively. The whole process lasted for 15 days.

The experiment was divided into following groups: *E. coli* treated with Cd^{2+} alone (Cd-*E. coli*), *E. coli* treated with Mt alone (Mt-*E. coli*), *E. coli* treated with Mt and Cd^{2+} (Mt-Cd-*E. coli*), and *E. coli* just cultured in LB (Wild-*E. coli*).

Characterization methods and MIC analysis

The bacteria, mineral, or complex samples were collected by centrifugation after 15 days of Cd induction experiment. All samples were washed three times with DI water and then lyophilized for subsequent characterization such as transmission electron microscope (TEM), scanning electron microscopy (SEM), and Fourier transform infrared (FTIR). Detailed characterization and procedures are provided in the "Supplementary information." The bacterial cells used for MIC test after 15 days of Cd^{2+} induction experiments are shown in the "Supplementary information."

RNA-seq and transcriptomic analysis

Bacterial cells from Wild-*E. coli*, Cd-*E. coli*, and Mt-Cd-*E. coli* were centrifuged and used for the total RNA extraction after 15 days of Cd induction experiment. The RNA-seq platform was an Illumina HiSeq X Ten with PE150. Raw sequences were quality-controlled and filtered by Trimmomatic and mapped to the *E. coli* ATCC 25,922 genome sequences (https://www.ncbi.nlm.nih.gov/nuccore/NZ_CP009072.1#) by Rockhooper2. All RNA sequencing data have been stored in the SRA (Sequence Read Archive) database of the National Center for Biotechnology Information (NCBI), which could be obtained by accession number (SRP263444). Differentially expressed genes (DEGs) were analyzed by comparing Cd-*E. coli* and Mt-Cd-*E. coli* with Wild-*E. coli*. The threshold for the DEGs was set as absolute values of $\log_2$ (fold change) ≥ 1 and *P* value or false discovery rate (FDR) < 0.05. The DEGs were subjected to biological functional analyses of Kyoto Encyclopedia of Genes and Genomes (KEGG) and Gene Ontology (GO).

qRT-PCR verification analysis

The samples of quantitative real-time PCR (qRT-PCR) were taken from RNA-seq samples to further verify the results of RNA-sequencing (RNA-seq). Nine ARGs (*mrcA*, *suhB*, *tktA*, *ydhC*, *marR*, *marB*, *ybjG*, *rplD*, and *uppP*) with significant changes were used for qRT-PCR verification according to the method of Xu et al. (2016). The internal reference gene was 16S rRNA. The primers used in the research are shown in Table S1, in the "Supplementary information." The $2^{-\Delta\Delta Ct}$ method was used to express the results connected with the internal reference gene expression in each sample (Schmittgen and Livak 2008). RNA-seq and qRT-PCR were conducted by OE Biotech Co., Ltd. (Shanghai, China).

Statistical analysis

All experiments were repeated independently in triplicate. The data shown in the corresponding figures and tables were the mean values of experiments. Besides, the data shown in Fig. 3 were expressed as mean ± standard deviation (s.d.). Statistical analyses were performed using the software OriginPro version 8.5 (OriginLab, USA).

Results

The XRD of Mt, Mt-Cd, Mt-E.coli, and Mt-Cd- E.coli

XRD of Mt, Mt-Cd, Mt-*E.coli*, and Mt-Cd-*E.coli* are shown in Fig. 1. The d_{001} value of Mt and Mt-*E.coli* was 1.57 nm and 1.59 nm, respectively. This indicates that the interactions between bacteria and Mt did not change the layer structure of Mt. However, the d_{001} value of Mt-Cd and Mt-Cd-*E.coli* was 1.81 nm and 1.67 nm, respectively. It was indicated that Cd^{2+} could enter into the interlayer of Mt and increase the interlayer spacing of Mt, and this also illustrated that the adsorption of Cd^{2+} on Mt was achieved by ion exchange. It was noteworthy that new diffraction reflection at 11.8° appeared in patterns of Mt- *E.coli* and Mt-Cd- *E.coli*, which implied a phase transition of the minerals after interacting with bacteria (Ruan et al., 2018). The results showed that there were some direct (e.g., phase transition of the minerals)

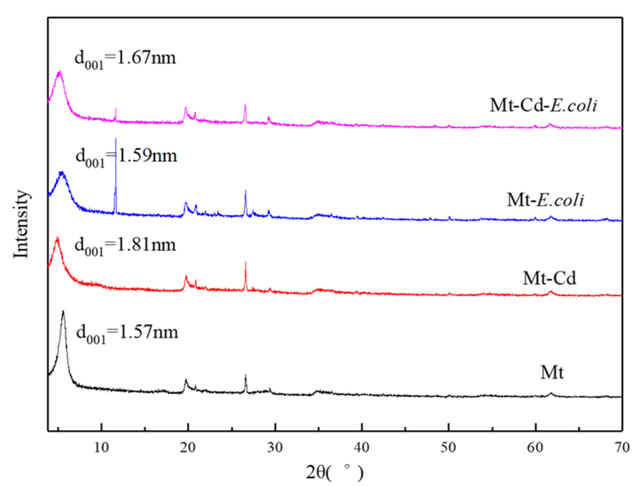

Fig. 1 XRD patterns of solid products: Mt, Mt–Cd, Mt-*E.coli*, and Mt-Cd- *E.coli*

or indirect (e.g., Mt adsorbed Cd^{2+} then indirectly influenced bacteria) interactions between bacteria and clays.

Adsorption of Cd by Mt, E. coli, and Mt-E. coli

Adsorption of Cd by Mt, *E. coli*, and Mt-*E. coli* is shown in Fig. 2. Mt-*E. coli* complex system had a high Cd adsorption capacity (67.6–82.1%), which is attributed to the adsorption properties of bacteria and Mt. Mt was found with a good adsorption capacity towards Cd^{2+} (60–67.2%). Bacteria was found with gradual increase in adsorption capacity (−2.57 to 5.94%) during Cd^{2+} induction, indicating that bacteria have enhanced resistance by developing their metabolic activity to resist Cd^{2+} stress. The negative absorption rate at low concentration may be attributed to the release of acidic substances by *E. coli* that may lead to the dissolution of Cd^{2+} in NB. Besides, the adsorption capacity of Mt-*E. coli* complex was higher than the sum of the individual Mt and *E. coli*. The reason may be the interfacial interactions between bacteria and Mt that changed the surface properties of them, improving their adsorption capacity towards Cd^{2+}. It was previously reported that soil bacteria and clays interacted with each other and their surface hydrophobicity, surface charge, and even surface structure might change, thus affecting the original adsorption behavior (Ruan et al., 2018).

Effect of Mt on antibiotic resistance in E. coli induced by Cd

Four different classes of antibiotics were selected. They were ampicillin (belonging to semisynthetic penicillin), tetracycline (belonging to tetracycline antibiotic), erythromycin (belonging to macrolide antibiotic), and chloramphenicol (belonging to amide alcohol antibiotic). According to Table 1, the MICs of Wild-*E. coli*/Cd-*E. coli*/Mt-Cd-*E. coli*/Mt-*E. coli* against ampicillin, tetracycline, erythromycin and chloramphenicol were 1/4/2/1 μg•mL^{-1}, 1/8/4/1 μg•mL^{-1}, 2/4/4/2 μg•mL^{-1}, and 2/4/2/2 μg•mL^{-1}, respectively. Compared to other three control groups (Wild-*E. coli* /Mt-Cd-*E. coli*/ Mt-*E. coli*), the MIC values of Cd-*E. coli* group against the four antibiotics (ampicillin, tetracycline, erythromycin, and chloramphenicol) were the highest, indicating that *E. coli* was more resistant to antibiotic under the induced by Cd^{2+}. The results of MIC demonstrated that Cd^{2+} could promote antibiotic resistance of *E. coli*, Mt could weaken the promotion of resistance under Cd^{2+}, and Mt alone did not change resistance of bacteria. It should be noted that the added MIC values of erythromycin in Cd-*E.coli* and Mt-Cd-*E.coli* and chloramphenicol in Mt-Cd-*E.coli* and Mt-*E. coli* were the same, which might be due to tiny differences in resistance, resulting in unremarkable difference in MIC data.

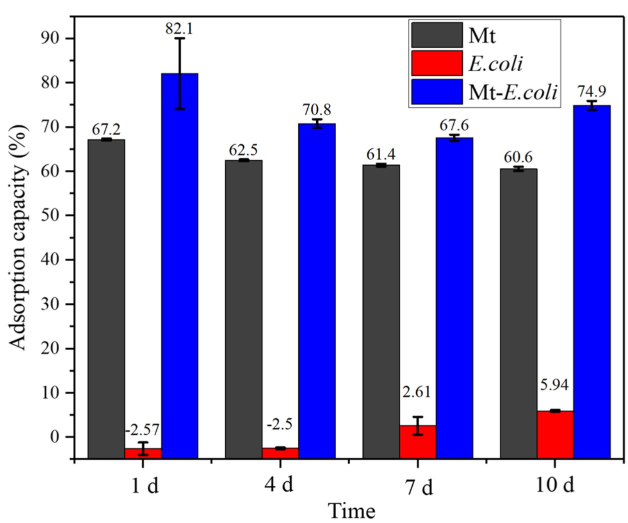

Fig. 2 Cd adsorption capacity of Mt, *E. coli* and Mt- *E. coli* in the culture media ((1, 4, 7, and 10 d represent the time after Cd induction at 16, 32, 64 and 128 mg L^{-1} for 24 h, respectively; pH, 7.0; Mt dosage, 16 g L^{-1})

The effect of Mt on resistant resistance under Cd stress

Genotype determines the phenotype. In order to further explore the resistance difference of *E. coli* between different treatment groups, RNA-seq and qRT-PCR detection were used.

Analysis of RNA-seq data

Gene expression of different groups was analyzed by RNA-seq. Venn diagram of DEGs (Fig. 3) showed that there were significant differences in gene expression between different treatment groups. Compared to Wild-*E. coli*, Cd-*E. coli* and Mt-Cd-*E. coli* shared 464 DEGs. Among them, 184/270 DEGs were up-regulated/down-regulated in

Table 1 Comparison of MIC before induction in Wild-*E. coli* and after it in Cd-*E. coli* and Mt-Cd-*E. coli*

Antibiotic	Wild-*E. coli*	Cd-*E. coli*	Mt-Cd-*E. coli*	Mt-*E. coli*
Ampicillin (μg•mL^{-1})	1	4	2	1
Tetracycline (μg•mL^{-1})	1	8	4	1
Erythromycin (μg•mL^{-1})	2	4	4	2
Chloramphenicol (μg•mL^{-1})	2	4	2	2

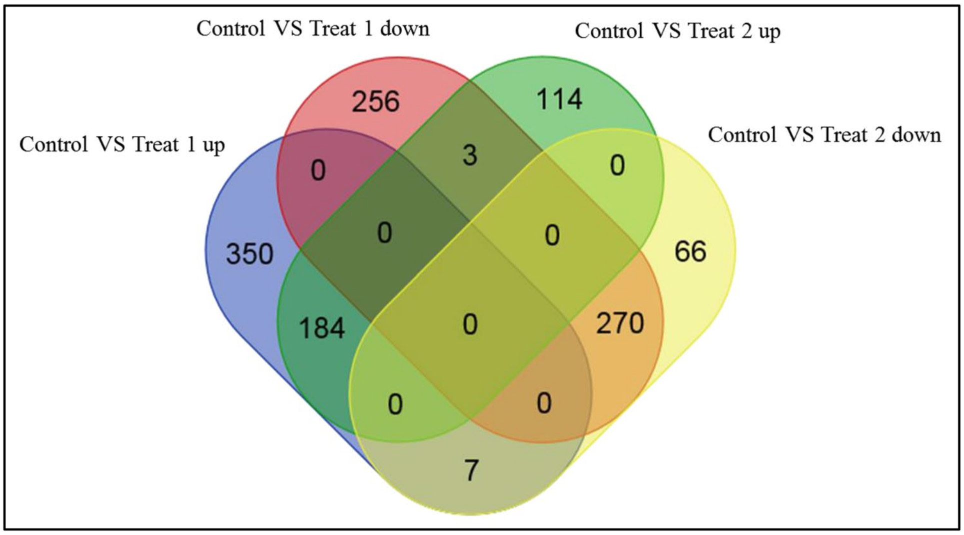

Fig. 3 Venn diagram of DEGs from comparison Treat 1 (Cd-*E. coli*) and Treat 2 (Mt-Cd-*E. coli*) with control (Wild-*E. coli*)

Cd-*E. coli* and Mt-Cd-*E. coli*, 7 DEGs were up-regulated in Cd-*E. coli* but down-regulated in Mt-Cd-*E. coli*, and 3 DEGs were up-regulated in Mt-Cd-*E. coli* but down-regulated in Cd-*E. coli*. Besides, 350/114 DEGs were up-regulated and 256/66 DEGs were down-regulated only in Cd-*E. coli*/Mt-Cd-*E. coli*, respectively. What is more, Fig. S1 (in the Supplementary information) visually reflected the significant difference between Mt-Cd-*E. coli* and Cd-*E. coli*.

GO and KEGG analysis on above DEGs were used to further elucidate the unique regulatory role of Mt in the process of developing antibiotic resistance under Cd^{2+} stress. GO functional analysis is shown in Fig. S2, in the Supplementary information. The biological process of DEGs was mainly reflected in cellular process, metabolic process, response to stimulus, localization, and biological regulation. The cellular component of DEGs was mainly reflected in cells, cell part, membrane, membrane part, macromolecule complexes, etc. Particularly, virion was only detected in Cd-*E. coli*. The molecular function of DEGs was mainly reflected in binding, catalytic activity, transporter activity, structural molecule activity, molecular transducer activity, etc. Particularly, electron carrier activity was only detected in Mt-Cd-*E. coli*, while protein binding transcription factor activity and translation regulator activity were only detected in Cd-*E. coli*. Furthermore, KEGG pathway analysis (Fig. S3, in the Supplementary information) indicated that the bacterial life activities in different groups were significantly different. Notably, immune system, cell growth and death were only detected in Cd-*E. coli*. The pathways associated with metabolism and environmental information processing of *E. coli* were most active under Cd^{2+} stress, suggesting that the two processes could provide a strong basis for further study on the regulatory mechanism.

Induction effects of Cd on ARGs and inhibition effects of Mt on these ARGs

RNA-seq results confirmed that the ARG expression of *E. coli* under Cd^{2+} stress with or without Mt was significantly different, as shown in Fig. 4. Compared to Wild-*E. coli*, Cd-*E. coli* increased bacterial expression of genes related to penicillin (e.g., *mrdA*, *mrcA*, and *dacB/D*), tetracycline (e.g., *accC*), macrolide (e.g., *macB*), streptomycin (e.g., *suhB*), and chloramphenicol (e.g., *tktA*, Table S2, in the Supplementary information). Moreover, compared to Wild-*E. coli*, the expression of genes in Cd-*E. coli* was up-regulated. These genes were related to multidrug resistance/multiple antibiotic resistance/response to antibiotic (e.g., *mdtQ*, *marB/R*, *arnA/B/C/D*) and multidrug efflux (e.g., *emrA*, *mprA*, *ydhC*, *marA*, Table S2, in the Supplementary information). This indicated that Cd could significantly increase the expression of the above genes, promoting antibiotic resistance directly (up-regulated gene expression in multidrug efflux conferred resistance by transporting antibiotics) or indirectly (up-regulated gene expression in penicillin binding conferred resistance by reducing binding of the antibiotic to the active site). However, compared to Cd-*E. coli*, Mt-Cd-*E. coli* reduced the expression of these genes to varying degree, inhibiting the promotion of antibiotic resistance.

qRT-PCR validation analysis

To evaluate the accuracy of gene expression levels in RNA-seq results, 9 Cd-responsive ARGs were verified by qRT-PCR. Cd-responsive ARGs were verified by qRT-PCR to evaluate the accuracy of gene expression levels in RNA-seq results. Figure 5 shows that the qRT-PCR detection results of all the selected genes were basically consistent

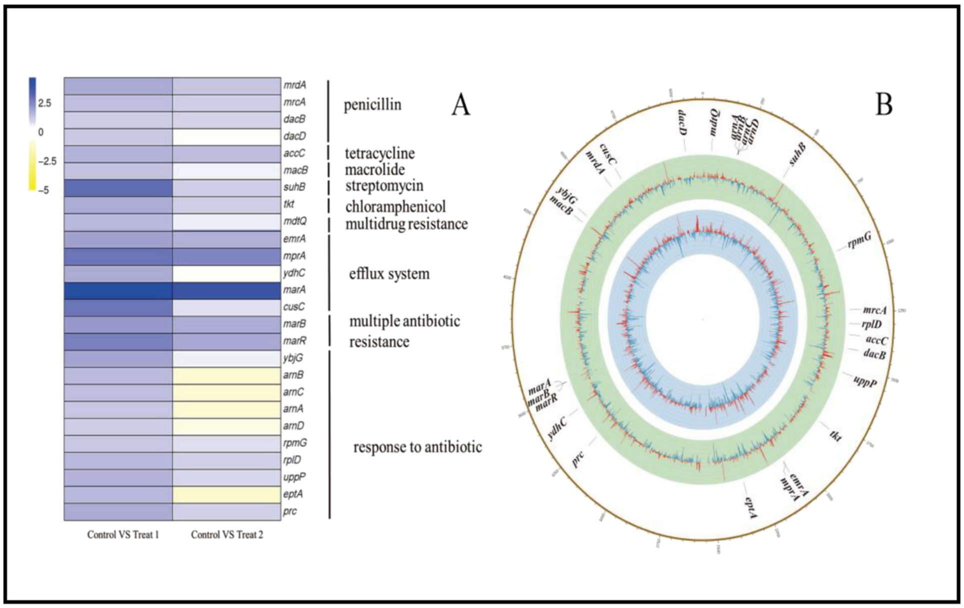

Fig. 4 **A** Differential expression of ARGs from comparison Treat 1 (Cd-*E. coli*) and Treat 2 (Mt-Cd-*E. coli*) with control (Wild-*E. coli*). Each gene was represented by a comparison of $\log_2$ (Fold Change) between the gene FPKM of Cd-*E. coli*, Mt-Cd-*E. coli* and Wild-*E. coli*. Blue represented up-regulated gene expression, and orange represented down-regulated gene expression. **B** Global analysis of transcript levels in Cd-*E. coli,* Mt-Cd-*E. coli*, and Wild-*E. coli* by RNA-seq. From inside out, the blue circle corresponded to the difference comparing Cd-*E. coli* with Wild-*E. coli*, and the green circle corresponded to the difference comparing Mt-Cd-*E. coli* with Wild-*E. coli*. The red lines in each circle represented up-regulation of gene expression (fold change > 1), while blue represented down regulation of gene expression (fold change < − 1). The outermost circle represented the full-strain *E. coli* ATCC 25,922 genome. The Cd-responsive ARGs were marked at the appropriate genomic position

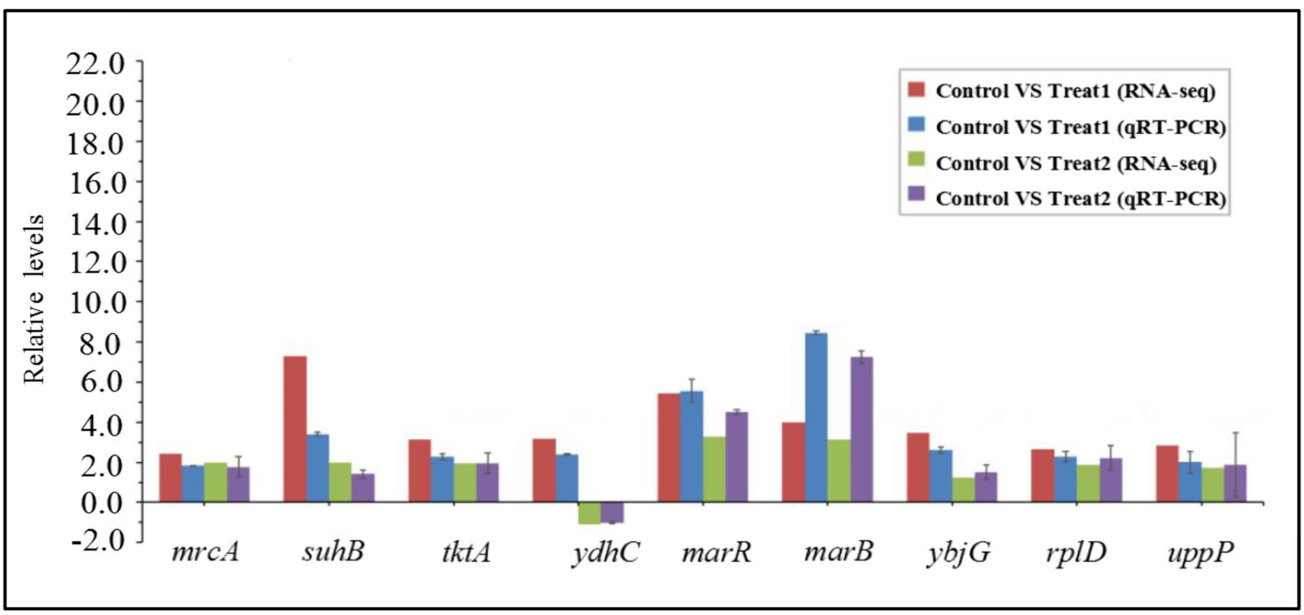

Fig. 5 Relative expressions of the selected antibiotics resistance gene. Comparison of the $\log_2$ (fold changes) between qRT-PCR and RNA-Seq data after Treat 1 (Cd-*E. coli*) and Treat 2 (Mt-Cd-*E. coli*) compared to control (Wild-*E. coli*)

with the RNA-seq detection results. It was reflected in that the expression levels of these genes were significantly up-regulated after Cd^{2+} treatment, and the up-regulated level was higher than that of Mt-Cd treatment (except for down-regulated genes). The variation multiples of the expression obtained by the two sequencing methods were different to some extent. This might be caused by the algorithmic differences in the amount of expression between various technologies or the decrease of diffusion efficiency with the increase of PCR quantitative circulation. Ultimately, this result further verified that Mt could reduce the expression of ARGs under Cd^{2+} stress.

Regulatory mechanism of Mt on ARGs in E. coli induced by Cd

In order to further clarify the regulatory mechanism of Mt alleviating antibiotic resistance promotion in *E. coli* under Cd^{2+} stress, and identify the genes playing a key role, KEGG enrichment analysis was conducted from two dimensions: (1) DEGs in Cd-*E. coli* and Mt-Cd-*E. coli* were compared to the Wild-*E. coli*, as shown in Fig. 6A, and (2) specific DEGs were obtained by comparing Mt-Cd-*E. coli* with Cd-*E. coli*, as shown in Fig. 6B. Significant enrichment pathways only existed in Fig. 6A indicating that these pathways played an important role under Cd^{2+} stress. Significant enrichment pathways only existed in Fig. 6B indicating the unique role of Mt. Significant enrichment pathways exist in Fig. 6A and Fig. 6B representing the prominent effects of Mt under Cd^{2+} stress. Combined with the above biological function analysis results, the key genes involved in the regulation of antibiotic resistance of *E. coli* under Cd^{2+} stress by Mt were discussed and analyzed in the following.

Protein synthesis

Pyrimidine/purine and ribosome played an important role in gene expression and protein synthesis of organisms, respectively. Ultimately, they resisted external toxicity by synthesizing related peptides or proteins. Figure 6A shows that ribosome, purine metabolism, and pyrimidine metabolism pathways were significantly enriched in common DEGs. Tetracyclines, macrolides, aminoglycosides and chloramphenicol could combine with the 30S subunits, 50S subunits, 30S subunits, and 50S subunits of bacterial intracellular ribosomes, respectively, contributing to the interference of protein synthesis. RNA-seq result showed that Cd-*E. coli* significantly increased the expression of genes related to 30 S ribosomal protein (such as, *rpsI/J/O/P/R/T/U*), 50 S ribosomal protein (such as *rplY/B/C/D/W/M/U/S/K/A/J/L*, *rpmB/G/H*, Table S2, in the Supplementary information), Mt-Cd-*E. coli* also increased the expression of some of these genes, but the expression levels were lower than that of Cd-*E. coli*. Besides, the up-regulated levels of genes related to pyrimidine production (e.g., *psuG*, *pyrE*, *thyA*) and *adk* gene (Table S2, in the Supplementary information) were higher in Cd-*E. coli* than that in Mt-Cd-*E. coli*. These genes catalyzed the interconversion of adenosine nucleotides and played an important role in maintaining cell stability, respectively. This suggested that the defense mechanism of bacterial protein synthesis was activated by Cd^{2+} stress, increasing the resistance to antibiotics. It was obvious that the inhibition effect of Mt on the up-regulated expression of genes related to protein synthesis process reduced the promotion of antibiotic resistance.

Transport system

The multi-drug resistance (MDR) pump and the major facilitator superfamily (MFS) protein family in bacteria transported heavy metals and antibiotics out of the cell, and *DsbA-DsbB* system was closely connected with metal efflux and the formation of multi-drug resistance system (Hayashi et al. 2000). In this study, the up-regulated levels of *marA* and *emrA* genes related to MDR efflux pump and multi-drug efflux MFS transporter were higher in Cd-*E. coli* than that in Mt-Cd-*E. coli* (Table S2, in the Supplementary information). The *DsbA* gene associated with disulfide exchange protein *DsbA* was only significantly up-regulated in Cd-*E. coli* (Table S2, in the Supplementary information). This suggested that Cd^{2+} activated the cross-resistance mechanism in *E. coli*, which enhanced the ability of antibiotic efflux, increasing their antibiotic resistance. Besides, *TatC* was an essential component of *Tat* transporter enzyme in *E. coli*, which was an important intracellular protein transport system. The *tatC* gene related to the synthesis of protein translocase subunit *TatC* was up-regulated by 1.75 and 1.04 in Cd-*E. coli* and Mt-Cd-*E. coli*, respectively, which might also be related to the cross-resistance in *E. coli.* Furthermore, ABC transporters played an important role in the detoxification and resistance of organisms. It could be seen that ABC transport was only shown in Fig. 6A. The genes related to ABC transport were significantly down-regulated in Cd-*E. coli* and Mt-Cd-*E. coli* (e.g., *hisM*, *yhdX/Y/W*, Table S2, in the Supplementary information), and Cd-*E. coli* showed more down-regulated expressions. This suggested that the up-regulated expression of ARGs in bacteria was induced by enhancing toxic effect of intracellular Cd^{2+}, which was due to the inhibition of ABC transport activity. Ultimately, antibiotic resistance was increased. The difference of gene expression was a vital reason for the difference of antibiotic resistance. Therefore, inhibition of the up-regulated and down-regulated effects of Mt on genes related to the transport system was a potential mechanism for weakening antibiotic resistance.

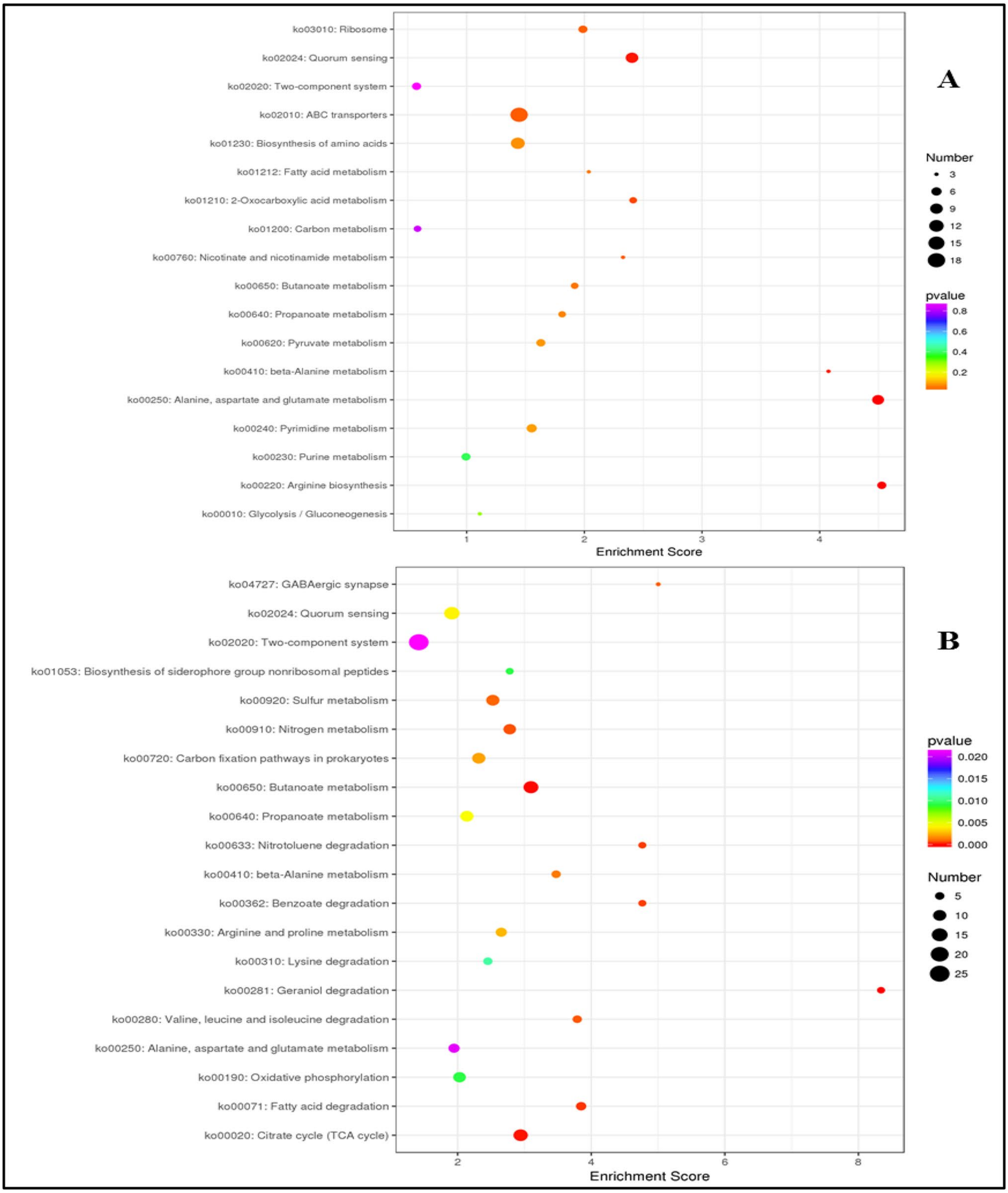

Fig. 6 **A** The KEGG enrichment top 20 plot for the common DEGs in Cd-*E. coli* and Mt-Cd-*E. coli* compared with Wild-*E. coli*. It was to explore the 20 pathways with the most significant differences among the common DEGs. **B** The KEGG enrichment top 20 plot for the specific DEGs in Mt-Cd-*E. coli* compared with Cd-*E. coli*. It was to explore the 20 pathways with the most significant differences among the specific DEGs. In the figure, the X-axis represented the Enrichment Score. Items with larger bubbles contained more differentially expressed genes. The smaller the enrichment *p* value of the bubble color from purple to red, the greater the significance

Signal transduction

Co-regulation was the process in which a series of transcriptional and translational response systems in bacteria respond to metal or antibiotic stress. For example, two-component system enabled bacteria to perceive, respond, and adapt to environmental changes through a variety of signal transduction processes (Wuichet et al. 2010). ArcA was a transcription factor of two-component system containing sensor domain and DNA-binding domain (Kohanski et al. 2010). After Cd^{2+} treatment, the expression of *acrR* gene decreased, which increased the transcription of *marA* gene related to encoding AcrAB in MDR effluent pump, resulting in antibiotic cross-resistance (Table S2, in the Supplementary information) (Randall and Woodward 2002). This suggested that Cd^{2+} activated the operation of intracellular co-regulation mechanism, and bacteria enhanced the resistance to antibiotics through the stress response of two-component system. Two-component system pathway was significantly enriched in Fig. 6A and 6B . Differences in the expression of above genes in Cd-*E. coli* and Mt-Cd-*E. coli* suggested that Mt's mitigation effect on gene expression related to the two-component system was an important mechanism for weakening antibiotic resistance. Quorum sensing meant that microbes sensed signals and made a coordinated response to external pressure by mutual communication (Deng et al. 2011). The *qseB* gene related to QseB was only significantly up-regulated in Cd-*E. coli* (Table S2, in the Supplementary information), indicating that *E. coli* could use quorum sensing regulator B (QseB) to regulate two-component system under Cd stress (Sperandio et al. 2010). The *yidC* gene related to the membrane protein insertion enzyme YidC was only significantly up-regulated in Cd-*E. coli* (Table S2, in the Supplementary information). This suggested that under Cd^{2+} stress, *E. coli* could positively regulate expression of genes related to cell membrane function through quorum sensing signal, thus enhancing drug resistance (Mata et al. 2000). Communication between nerve immunity could affect the immune response by releasing neurotransmitters and other regulatory factors through synapses. γ-aminobutyric acid (GABA) was an inhibitory neurotransmitter regulated by phosphorylation mechanism. It could regulate the migration and transport of cell surface materials by signal transduction. The expression of *gadA* and *gadB* genes related to glutamate decarboxylase in Cd-*E. coli*/Mt-Cd-*E. coli* were 3.23/ − 1.04 and 3.15/ − 0.97, respectively (Table S2, in the Supplementary information). The *gadB* gene encoded a key enzyme in GABA synthesis pathway. Furthermore, DEGs associated with oxidative phosphorylation synthesis were only significantly down-regulated in Cd-*E. coli*, including the genes which encoded fumarate reductase (e.g. *frda*/*b*/*c*/*d*) and succinate dehydrogenase (e.g. *sdhA*, Table S2, in the supplementary information). This indicated that Cd^{2+} negatively regulated the expression of genes related to oxidative phosphorylation and positively induced the expression of genes related to GABAergic synapse. Finally, antibiotic resistance was increased by enhancing signal-mediated migration and transport. Compared to Cd-*E. coli*, the above DEGs showed inhibitory effect in Mt-Cd-*E. coli*; i.e., the up-regulated and down-regulated expression levels were reduced. Moreover, GABAergic synapse and oxidative phosphorylation pathways were only significantly enriched in Fig. 4B, indicating that the regulatory role of Mt in signal transduction was an important mechanism to alleviate the promotion of antibiotic resistance.

Cellular component

A large number of studies showed that biofilms could effectively enhance the resistance of metals and antibiotics, which could be used as their co-selection model (Baker-Austin et al. 2006). In Cd-*E. coli*, genes related to polysaccharide biosynthetic (such as *wcaA*/*B*/*C*/*D*/*F*/*I*/*J*/*K*/*L*) and *DR76_RS20260* gene related to biofilm regulatory protein *BssS* (Table S2, in the Supplementary information) were only significantly up-regulated in Cd-*E. coli*. The up-regulated levels of genes related to cell adhesion (e.g., *znuA*, *DR76_RS19090*, Table S2, in the Supplementary information) were higher in Mt-Cd-*E. coli* than that in Cd-*E. coli*. Therefore, heavy metals could stimulate the production of extracellular polymeric substances (EPS) in prokaryotes, so that organisms could develop cell adhesion or generate biofilms to regulate antibiotic resistance. Cells attached to clay surfaces could secrete more polymers in vitro to enhance adhesion (Lai et al. 2019). Therefore, Mt-Cd-*E. coli* had different resistance expression mechanisms. It was important to note that fatty acids were a vital component of phospholipids as well as of cell membranes. Figure 4 indicates that genes related to fatty acid metabolism were significantly different between the two treatment groups. Besides, penicillin could interfere with peptidoglycan synthesis by inhibiting penicillin binding proteins (PBPs) (Arabski et al. 2007). Genes annotated as peptidoglycan biosynthesis pathway (e.g., *mrdA*, *mrcA*, *dacB*/*D*, Table S2, in the Supplementary information) were only significantly up-regulated in Cd-*E. coli*, and these genes were closely related to the binding function of penicillin.

Energy supply

Energy was a basic need of life, which provided the necessary conditions for survival of organisms. From Fig. 6, it could be seen that citrate cycle (TCA cycle), nitrogen metabolism, sulfur metabolism, carbon fixation pathways in prokaryotes, pyruvate metabolism, and glycolysis/gluconeogenesis pathways involved in the energy metabolism process were significantly enriched. Alanine, aspartate, and glutamate metabolism, butanoate metabolism, propanoate metabolism, lysine degradation, valine, leucine, and isoleucine degradation pathways related to the TCA cycle were also significantly enriched, as shown in Fig. 6. This suggested that energy supply played an important role in the development of bacterial antibiotic resistance under Cd^{2+} stress. In particular, Mt made the gene expression difference more significant. For example, most of the DEGs related to TCA cycle, sulfur metabolism, carbon fixation pathways in prokaryotes, and glycolysis/gluconeogenesis pathways in Cd-*E. coli* were down-regulated. Besides, almost all of these genes in Mt-Cd-*E. coli* were significantly up-regulated compared to Cd-*E. coli*. This suggested that under Cd stress, bacteria were in a more "energy storage" state by inhibiting these pathways. As a result, their antibiotic resistance was enhanced. The nutrients and buffering provided by Mt enable the bacteria to have different strain patterns in a more "energy-consuming" state (Warr et al. 2009).

Discussion

In this manuscript, we introduce the potential roles of Mt in the emergence of ARGs under Cd^{2+} stress. Mt interacted with *E. coli* under Cd^{2+} stress, resulting in significantly different gene expression between Cd-*E. coli* and Mt-Cd-*E. coli*, which formed a unique mode of operation in the presence of Mt. Therefore, it is necessary to provide a new idea for alleviating the promotion of antibiotic resistance under Cd^{2+} stress, and theoretical basis for the research on environmental pressure and ecological risks caused by the emergence of antibiotic resistance. This study has confirmed the inhibition effect of Mt on ARGs in *E. coli* induced by Cd^{2+} and the regulatory mechanism at the molecular level. The characterization methods of TEM, SEM, and FTIR were used to further discuss this regulatory effect in the following.

TEM observations showed that normal Wild-*E. coli* had intact cell walls and intracellular material (Fig. 7A). Cd-*E. coli* was damaged to a certain extent, which was manifested in the thinning of cell wall and significant autolysis. What is more, pores were formed on the cell surface and more granular sediments were contained in the cell (Fig. 7B). The cell wall of Mt-Cd-*E. coli* was thinner, but the autolysis was

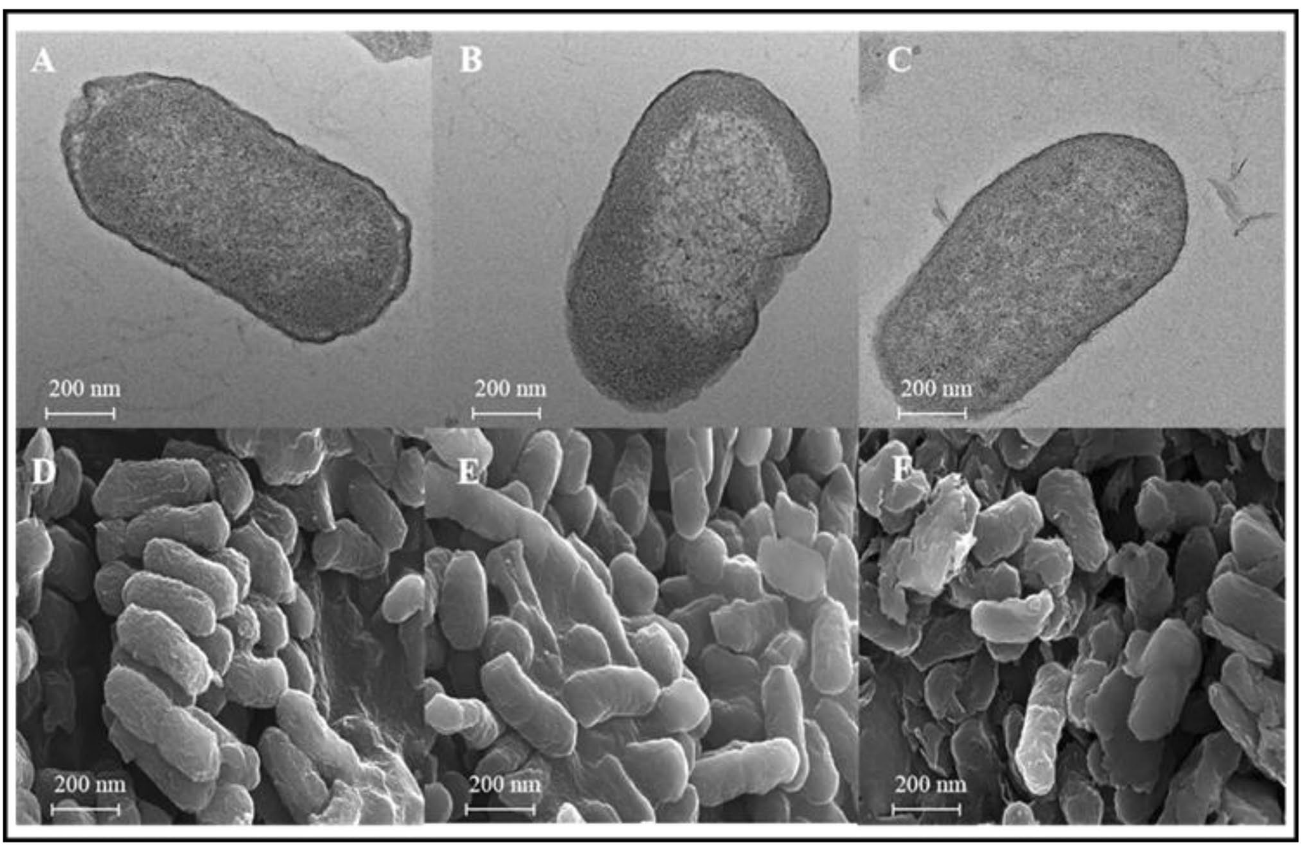

Fig. 7 TEM images of **A** Wild-*E. coli*, **B** Cd-*E. coli*, **C** Mt-Cd-*E. coli*, and SEM images of **D** Wild-*E. coli*, **E** Cd-*E. coli*, **F** Mt-Cd-*E. coli*

not significant. Besides, its pore formation was less than Cd-*E. coli* (Fig. 7C). Thus, *E. coli* maintained a relatively complete morphology under the stress of Cd in the presence of Mt. *E. coli* had non-specific pore proteins *OmpC* and *OmpF* on the extracellular membrane whose syntheses were regulated by *ompB3* gene (Torres et al. 2006; Tokunaga et al. 2013). RNA-seq data showed that the expression of *ompC* and *ompB* genes in Cd-*E. coli*/Mt-Cd-*E. coli* were 1.75/1.16 and 1.02/0.53, respectively. Combined with gene expression of the transport system, it was observed that Cd-*E. coli* enhanced antibiotic resistance by forming cell membrane pores and stimulating the transport system to promote the efflux of intracellular substances. Mt had a protective effect on bacterial morphology under Cd^{2+} stress, which reduced the formation of pores and affected transport system to some extent, inhibiting the promotion of antibiotic resistance.

As depicted in SEM images, the normal Wild-*E. coli* was rod-shaped and had a bumpy surface (Fig. 7D). Cd-*E. coli* had a smooth surface (Fig. 7E), suggesting that the addition of Cd changed the cell's surface morphology. Mt-Cd-*E. coli* adhered to the Mt surface. The morphology of Mt-Cd-*E. coli* also tended to be smooth, but was different from that of Cd-*E. coli* (Fig. 7F). The smoother surface of bacteria under Cd^{2+} stress was related to the Cd^{2+} stimulating bacteria to produce EPS. Besides, EPS could significantly enhance antibiotic resistance (Baker-Austin et al. 2006). Therefore, under Cd stress, *E. coli* produced EPS to resist the adverse environment, thus affecting antibiotic resistance. Study showed that EPS rich in macromolecular polysaccharides played an important role in the adhesion between bacteria and matrix (More et al. 2014), which resulted in the difference of above gene expression.

The FTIR spectra of Mt, Wild-*E. coli*, Cd-*E. coli* and Mt-Cd-*E. coli* are shown in Fig. 8. It was observed that

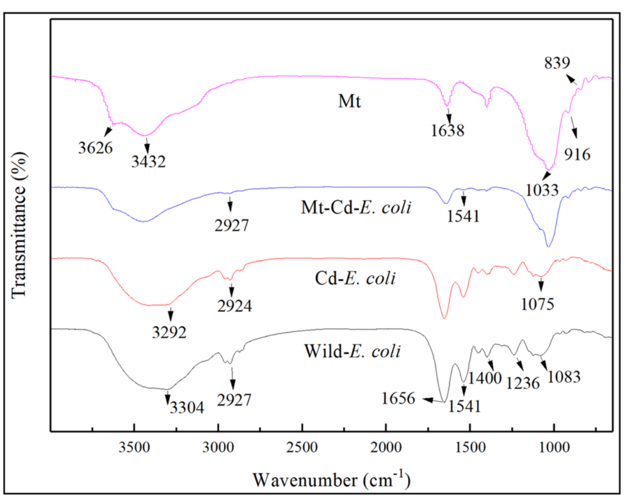

Fig. 8 FTIR spectra of Wild-*E. coli*, Cd-*E. coli*, Mt-Cd-*E. coli*, and Mt

some characteristic peaks related to *E. coli* changed after Cd^{2+} or Mt-Cd treatment. The peaks at 3304, 2927, 1656, 1541, 1400, 1236, and 1083 cm^{-1} represented stretching vibration of hydroxyl and amine groups in proteins, -CH stretching vibration in alkyl chain, C = O stretching vibration in amide I, -NH bending vibration and C-N stretching vibration in amide I, vibration peak of carboxylate ion, $-SO_3$ stretching vibration, and -CN stretching vibration in the protein component, respectively (Chen et al. 2012; Pei et al. 2009; Tunali et al. 2005). These functional groups were closely related to cell wall and membrane. Changes in cell wall or membrane composition could enhance antibiotic resistance (Baker-Austin et al. 2006; Sieradzki et al. 2008). Compared to Wild-*E. coli*, these characteristic peaks of Cd-*E. coli* shifted to the lower wavenumbers or weakened, and the peak intensity decreased or disappeared in Mt-Cd-*E. coli.* It indicated that the composition of cell wall and membrane changed under Cd^{2+} stress, which affected antibiotic resistance. Compared to Mt, new characteristic peaks appeared at 2927 and 1541 cm^{-1} in Mt-Cd-*E. coli.* It suggested that aliphatic compounds and proteins on the cell membrane surface of bacteria could interact with active sites on mineral surfaces (Holmes et al., 2009), causing differences in antibiotic resistance. This result was in consistent with the results of gene expression related to cellular component (Sect. 3.3.4). Besides, the changes in ARGs caused by exposure to montmorillonite probably due to the following two aspects. First, based on the structure of Mt, it has a big cation exchange capacity. Cd^{2+} can be exchanged with interlayer cations (Ca^{2+}, Mg^{2+}) of Mt. This can reduce the stress of Cd^{2+} to bacteria and the ARGs expression of bacteria. Second, Mt has a large specific surface area, which provides a good habitat for bacteria to survive. Meanwhile, Mt can protect the clay-bound DNA or extracellular polymeric substances (EPS) from Cd^{2+}. Furthermore, serving as a physical barrier for external protection, EPS can also provide functional group active sites that interact with Cd to reduce the risk of exposure to ARGs.

In summary, from the perspective of genes and phenotypes related to antibiotic resistance, it has been confirmed that Cd^{2+} could induce antibiotic resistances of *E. coli* and Mt could inhibit the promotion of these resistances. Mt interacted with *E. coli* under Cd stress, resulting in significantly different gene expression between Cd-*E. coli* and Mt-Cd-*E. coli*, which formed a unique mode of operation in the presence of Mt. Based on the cross-resistance and co-regulation mechanism of *E. coli* between heavy metals and antibiotics, the regulatory mechanism of Mt on antibiotic resistance in *E. coli* induced by Cd^{2+} was elucidated. Therefore, this study not only provided a new idea for alleviating the promotion of antibiotic resistance under Cd^{2+} stress, but also provided scientific support and theoretical basis for the research on environmental pressure and ecological risks caused by the emergence of antibiotic resistance.

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Author contribution Yihao Li and Huimin Wang conducted the experiments and wrote the manuscript. Pingxiao Wu: Conceptualization; supervision; project administration; funding acquisition. Lu Jiang and Zubair Ahmed: Writing—review and editing. Bo Ruan: Data curation. Shanshan Yang: Conceptualization.

Data availability All RNA sequencing data have been stored in the SRA (Sequence Read Archive) database of National Center for Biotechnology Information (NCBI), which could be obtained by Accession Number (SRP263444). The data sets generated during and/or analyzed during the current study are either shown in the manuscript or available from the corresponding author on reasonable request.

Declarations

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare no competing interests.

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